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Comprehensive Question Bank: Chapter 7 — Hypothesis Testing

Section A: Multiple-Choice Questions (MCQs)

Topic 7.1: Introduction to Hypothesis Testing

1. What is a hypothesis?
   A) A final, unquestionable result
   B) A testable, measurable assumption or statement
   C) A type of dataset
   D) A programming function

2. Hypothesis testing is best described as:
   A) A method to store large datasets
   B) A statistical method used to make decisions using data
   C) A way to visualize data only
   D) A programming language feature

3. Why is hypothesis testing important in data analysis?
   A) It eliminates the need for data collection
   B) It guarantees 100% accurate conclusions
   C) It supports evidence-based decisions and reduces guesswork
   D) It replaces the need for a research question

4. A research question is best described as:
   A) A final answer to a study
   B) The main question that defines what is being studied
   C) A type of test statistic
   D) A dataset used for testing

5. Which of the following is a property of a well-formed hypothesis?
   A) It is vague and open to interpretation
   B) It is clear, measurable, and testable
   C) It cannot be tested with data
   D) It only applies to non-technical topics

6. In computer science, which of the following is an example of a valid hypothesis?
   A) "Our app looks nice"
   B) "The new caching algorithm reduces average response time"
   C) "Developers like coding"
   D) "The database is important"
7. What happens when a research question is converted into a hypothesis?
   A) The question becomes untestable
   B) It becomes a specific, measurable statement that can be checked with data
   C) It becomes purely a personal opinion
   D) It removes the need for any data collection
8. Which of the following best explains the role of hypothesis testing in software engineering?
   A) It replaces the need for software testing
   B) It helps determine whether a change (e.g., a new algorithm) has a real effect, not just random variation
   C) It is only used for hardware design
   D) It has no real application in engineering
9. A hypothesis predicts:
   A) Nothing measurable
   B) An outcome that can be checked using data
   C) Only qualitative opinions
   D) The final grade of a student
10. Which statement about hypothesis testing is TRUE?
   A) It guarantees the hypothesis is true
   B) It uses collected data to accept or reject an assumption
   C) It is unrelated to research questions
   D) It cannot be applied to A/B testing

Topic 7.2: Types of Hypotheses

1. What does the null hypothesis (H_0) represent?
   A) A claim of significant change
   B) The default assumption of no effect or no difference
   C) The final conclusion of a study
   D) A type of test statistic

2. What does the alternative hypothesis (H_1 or H_a) represent?
   A) No effect or no difference
   B) A claim that there is an effect or difference
   C) A dataset used for testing
   D) The significance level

3. In the courtroom analogy, the null hypothesis (H_0) is comparable to:
   A) The verdict of "guilty"
   B) The defendant being presumed innocent
   C) The judge's final decision
   D) The jury's opinion

4. Which symbol is commonly used to represent the null hypothesis?
   A) H_1
   B) H_a
   C) H_0
   D) H_2

5. A one-tailed hypothesis test is used when:
   A) You expect no difference at all
   B) You expect a difference in one specific direction (e.g., greater than or less than)
   C) You are testing categorical data only
   D) You are only visualizing data

6. A two-tailed hypothesis test is used when:
   A) You expect the result only to increase
   B) You expect the result only to decrease
   C) You are checking for a difference in either direction, without specifying which
   D) You do not need a null hypothesis

7. Which of the following is a correctly formulated null hypothesis for testing if a new AI code assistant reduces coding time?
   A) H_0: The AI assistant makes coding faster
   B) H_0: The AI assistant does not reduce coding time
   C) H_0: The AI assistant is popular among developers
   D) H_0: The AI assistant improves code quality

8. Which rule is essential when formulating H_0 and H_1?
   A) They should overlap to cover uncertain cases
   B) They must cover all possibilities and never overlap
   C) They should both claim an effect exists
   D) They do not need to be measurable

9. In statistics, we generally aim to:
   A) Prove H_1 is true with 100% certainty
   B) Gather enough evidence to reject H_0
   C) Ignore H_0 completely
   D) Avoid stating H_1 at all
10. Which of the following is an example of an alternative hypothesis?
   A) H_0: μ_A = μ_B
   B) H_1: μ_A ≠ μ_B
   C) α = 0.05
   D) p-value = 0.03
11. A hypothesis must be "falsifiable." This means:
   A) It must always be proven true
   B) It must be possible for data to prove it wrong
   C) It cannot be tested with data
   D) It applies only to qualitative research

Topic 7.3: Concepts in Hypothesis Testing

1. What is a test statistic?
   A) A fixed constant used in every test
   B) A numerical value calculated from sample data used to make a decision
   C) The final conclusion of a hypothesis test
   D) A type of graph
2. Which test statistic is commonly used for small sample sizes when the population standard deviation is unknown?
   A) z-test
   B) t-test
   C) F-test
   D) Chi-Square test
3. Who developed the t-distribution, and under what pseudonym was it published?
   A) Ronald Fisher, under "Researcher"
   B) William Sealy Gosset, under "Student"
   C) Karl Pearson, under "Analyst"
   D) Jerzy Neyman, under "Scientist"

4. The critical region in hypothesis testing is:
   A) The area where we accept H_0 without question
   B) The range of values where, if the test statistic falls there, H_0 is rejected
   C) A fixed numerical constant
   D) A chart used only for visualization
5. The significance level (α) represents:
   A) The probability that H_0 is true
   B) The maximum acceptable risk of wrongly rejecting a true H_0
   C) The sample size used in a test
   D) The test statistic value
6. A commonly used significance level in hypothesis testing is:
   A) 0.5
   B) 0.05
   C) 5.0
   D) 50
7. The p-value measures:
   A) The size of the sample
   B) The probability of obtaining the observed result (or more extreme) if H_0 is true
   C) The exact value of the population mean
   D) The number of variables in a study
8. If the p-value is smaller than the significance level (α), the correct decision is to:
   A) Accept H_0
   B) Reject H_0
   C) Ignore the result
   D) Increase the sample size only
9. A Type I Error occurs when:
   A) We fail to reject a false H_0
   B) We reject a true H_0 (a false alarm)
   C) We correctly reject H_0
   D) We calculate the wrong sample size
10. A Type II Error occurs when:
   A) We correctly reject H_0
   B) We fail to reject a false H_0 (a missed effect)
   C) We reject a true H_0
   D) We calculate an incorrect p-value

11. In the analogy comparing errors to a courtroom trial, a Type I Error is similar to:
    A) Letting a guilty person go free
    B) Sending an innocent person to prison
    C) A fair and correct verdict
    D) Ignoring the evidence completely
12. Degrees of freedom (df) for a simple one-sample t-test is calculated as:
    A) n
    B) n − 1
    C) n + 1
    D) 2n
13. A 95% confidence interval means:
    A) There is a 95% chance H_0 is true
    B) If the experiment were repeated many times, 95% of the calculated intervals would contain the true value
    C) The p-value is always 0.95
    D) The sample size must be exactly 95
14. What formula correctly represents the t-value?
    A) $t = (\mu - \bar{x}) \times s$
    B) $t = (\bar{x} - \mu) / (s / \sqrt{n})$
    C) $t = n / s$
    D) $t = (\bar{x} + \mu) \times n$
15. Choosing the significance level (α) should be done:
    A) After analyzing the results, to fit the data
    B) Before collecting or analyzing the data
    C) Only after calculating the p-value
    D) Only for Chi-Square tests

Topic 7.4: Performing Hypothesis Tests

1. What is the first step in performing a hypothesis test?
   A) Calculate the test statistic
   B) State the null and alternative hypotheses
   C) Set the significance level
   D) Visualize the data
2. What is the correct order of the standard hypothesis testing process?
   A) Collect data → Set significance level → State hypotheses → Calculate statistic
   B) State hypotheses → Set significance level → Collect data → Calculate statistic →

Make a decision

C) Make a decision → Collect data → State hypotheses

D) Calculate statistic → Collect data → State hypotheses

3. The F-test is primarily used to:
   A) Compare means between two groups
   B) Compare variances between two or more groups
   C) Test categorical/frequency data
   D) Visualize confidence intervals

4. The Chi-Square (χ^2) test is best suited for:
   A) Comparing variances of continuous data
   B) Testing categorical/frequency data for independence or goodness-of-fit
   C) Comparing two population means only
   D) Testing significance levels only

5. In a Chi-Square test, "O" and "E" represent:
   A) Optimal and Estimated values
   B) Observed and Expected frequencies
   C) Outlier and Error values
   D) Ordinal and Exponential values

6. The correct formula for the Chi-Square test statistic is:
   A) $\chi^2 = \Sigma(O + E)^2 / E$
   B) $\chi^2 = \Sigma(O - E)^2 / E$
   C) $\chi^2 = \Sigma(O \times E)^2$
   D) $\chi^2 = \Sigma(E - O) / O$

7. The correct formula for the F-test statistic is:
   A) $F = S1 + S2$
   B) $F = S1^2 / S2^2$
   C) $F = S1^2 \times S2^2$
   D) $F = S1 - S2$

8. In an A/B test comparing server speeds, which is the most appropriate null hypothesis?
   A) H_0: $\mu\_A > \mu\_B$
   B) H_0: $\mu\_A = \mu\_B$
   C) H_1: $\mu\_A = \mu\_B$
   D) H_0: $\mu\_A \neq \mu\_B$

9. Given a calculated $t = 2.5$ and a p-value of 0.02 with $\alpha = 0.05$, the correct decision is:
   A) Fail to reject H_0
   B) Reject H_0
   C) Recalculate the sample size
   D) Ignore the test statistic

10. When calculating expected frequencies in a Chi-Square goodness-of-fit test with equal expected proportions across 5 categories and a total of 100 observations, the expected frequency per category is:
   A) 100
   B) 20
   C) 5
   D) 50
11. If a calculated F-value is smaller than the critical F-value from the F-table, the correct decision is:
   A) Reject H_0; variances are significantly different
   B) Fail to reject H_0; no significant difference in variances
   C) Automatically reject H_1
   D) Recollect the data
12. Which Python library function is commonly used to perform a one-sample t-test?
   A) `numpy.mean()`
   B) `scipy.stats.ttest_1samp()`
   C) `pandas.describe()`
   D) `matplotlib.pyplot()`

Topic 7.5: Data Visualization For Hypothesis Testing

1. Why is data visualization important in hypothesis testing?
   A) It replaces the need for statistical tests
   B) It helps present data clearly and identify patterns before applying tests
   C) It only matters for non-technical audiences
   D) It eliminates the need for a hypothesis
2. Which chart type is best for comparing categories or groups?
   A) Line graph
   B) Bar chart
   C) Scatter plot
   D) Histogram (for a single distribution)
3. Which chart type is best suited for showing trends and changes over time?
   A) Pie chart
   B) Line graph
   C) Bar chart
   D) Box plot

4. Which chart type is most useful for displaying relationships between two continuous variables?
   A) Bar chart
   B) Pie chart
   C) Scatter plot
   D) Table
5. A box plot is especially useful for showing:
   A) Only the mean of a dataset
   B) The median, spread, and outliers of a dataset
   C) Only categorical frequencies
   D) The p-value directly
6. What can data visualization help identify in a dataset?
   A) Only the final p-value
   B) Outliers and anomalies
   C) The exact significance level
   D) The degrees of freedom only
7. If two groups' confidence intervals on a chart barely overlap, this visually suggests:
   A) There is definitely no difference
   B) There may be a meaningful difference between the groups
   C) The test must be invalid
   D) The sample size was too small to matter
8. Which of the following is a valid reason to be cautious when interpreting a bar chart alone?
   A) Bar charts are never useful
   B) A chart alone does not confirm statistical significance without the underlying test
   C) Bar charts always guarantee correct conclusions
   D) Charts cannot display averages

Topic 7.6: Ethical Issues in Data Science and Analysis

1. What is bias in data collection?
   A) Perfectly balanced and representative data
   B) An unfair or unrepresentative sample that skews results
   C) A type of test statistic
   D) A visualization technique
2. A survey about student performance conducted only in top-performing schools is an example of:
   A) Confirmation bias

B) Survey bias
C) Gender bias
D) Geographical bias

3. A health study that includes only young adults, ignoring older individuals, is an example of:
   A) Survey bias
   B) Sampling bias
   C) Confirmation bias
   D) Ethical data use
4. A hiring dataset containing mostly male candidates, which causes a model to favor male applicants, is an example of:
   A) Geographical bias
   B) Gender bias
   C) Sampling bias
   D) Survey bias
5. Using data only from urban areas to make decisions for rural populations is an example of:
   A) Gender bias
   B) Geographical bias
   C) Confirmation bias
   D) Survey bias
6. A researcher who only focuses on data supporting their existing belief while ignoring contradictory data is exhibiting:
   A) Survey bias
   B) Confirmation bias
   C) Sampling bias
   D) Ethical data use
7. What is "p-hacking"?
   A) A legitimate method of improving test accuracy
   B) Manipulating data or running many tests until a "significant" result appears by chance
   C) A method for visualizing p-values
   D) A way to increase sample size ethically
8. If 100 hypothesis tests are run on completely random, meaningless data at $\alpha = 0.05$, approximately how many tests are expected to show "significant" results purely by chance?
   A) 0
   B) 5
   C) 50
   D) 100

9. Which of the following is an ethical practice in data science?
   A) Reporting only the significant results out of many tests performed
   B) Reporting all tests conducted, including non-significant ones
   C) Choosing α after seeing the results
   D) Using biased samples for convenience

10. Ethical use of data requires:
    A) Ignoring user privacy for better results
    B) Respecting privacy, obtaining proper consent, and avoiding discrimination
    C) Using data for any purpose regardless of consent
    D) Hiding all limitations of a study

11. Which historical event is used to illustrate the danger of ignoring rigorous statistical analysis and communication?
    A) The invention of the t-test
    B) The Challenger Space Shuttle disaster
    C) Google's 41 shades of blue test
    D) The Lady Tasting Tea experiment

Topic 7.7: Communicating Results and Conclusions

1. If a hypothesis test results in "fail to reject H_0," the correct interpretation is:
   A) H_0 has been proven true with 100% certainty
   B) There is not enough evidence to reject H_0
   C) H_1 has been proven true
   D) The test was invalid

2. Which of the following is the correct way to phrase a hypothesis testing conclusion?
   A) "We proved H_1 is true."
   B) "We reject H_0 at the 0.05 significance level."
   C) "H_0 is 100% confirmed."
   D) "The test guarantees no error."

3. When presenting technical findings to a non-technical audience, it is best to:
   A) Use as much statistical jargon as possible
   B) Lead with a plain-English conclusion supported by clear visuals
   C) Avoid mentioning any limitations of the study
   D) Skip the original research question entirely

4. Why should a final conclusion always be connected back to the original research question?
   A) It is not necessary to do so
   B) It ensures the conclusion directly answers what was originally being investigated
   C) It replaces the need for a hypothesis
   D) It eliminates the need for data visualization

5. Which of the following best describes "responsible communication of findings"?
   A) Exaggerating small effects as major breakthroughs
   B) Presenting results honestly, including limitations, without exaggeration
   C) Hiding the sample size used
   D) Avoiding mention of the significance level

6. Misinterpreting test results can lead to:
   A) Always more accurate conclusions
   B) Incorrect conclusions and poor decision-making
   C) No consequences at all
   D) Better sample sizes automatically

7. Which statement correctly reflects statistical humility in interpreting results?
   A) A hypothesis test provides absolute proof
   B) A hypothesis test only provides evidence at a certain confidence level, not absolute proof
   C) A p-value guarantees the hypothesis is correct
   D) Statistical tests never have any risk of error

Section B: Short Answer Questions (SQs)

Topic 7.1: Introduction to Hypothesis Testing

1. Define a hypothesis in your own words.
2. What is a research question, and how does it differ from a hypothesis?
3. Explain briefly why hypothesis testing reduces guesswork in decision-making.
4. State three properties of a well-formulated hypothesis.
5. Explain briefly the role of hypothesis testing in modern software engineering.
6. Identify whether the following statement is a valid hypothesis: "Our app is good." Justify your answer.
7. Briefly explain how a vague research question can be transformed into a testable hypothesis, using an example.

Topic 7.2: Types of Hypotheses

1. Define the null hypothesis (H_0).
2. Define the alternative hypothesis (H_1 or H_a).
3. Differentiate between a one-tailed test and a two-tailed test.
4. Explain briefly why H_0 and H_1 must never overlap.
5. Using the courtroom analogy, explain the relationship between H_0, H_1, and the concept of "reasonable doubt."
6. Write a null hypothesis and an alternative hypothesis for testing whether a new database index reduces query time.
7. Explain briefly why a hypothesis must be "falsifiable."

Topic 7.3: Concepts in Hypothesis Testing

1. Define a test statistic.
2. Differentiate between a z-test and a t-test.
3. Explain briefly what the critical region represents in hypothesis testing.
4. Define the significance level (α) and state a commonly used value.
5. Explain what a p-value tells us, in plain words.
6. Differentiate between a Type I Error and a Type II Error.
7. State the decision rule used to compare the p-value with the significance level.
8. Define "degrees of freedom" and state the formula for a one-sample t-test.
9. Explain briefly what a 95% confidence interval means.
10. Identify the historical figure who developed the t-distribution and explain why he published under a pseudonym.

Topic 7.4: Performing Hypothesis Tests

1. List the five standard steps involved in performing a hypothesis test.
2. Differentiate between the F-test and the Chi-Square test in terms of their purpose.
3. Explain briefly what "Observed frequency" and "Expected frequency" mean in a Chi-Square test.
4. State the formula for the F-test statistic and explain each symbol.
5. Predict the decision (reject or fail to reject H_0) if a calculated p-value is 0.01 and $\alpha = 0.05$.
6. Briefly describe how an A/B test could be used to evaluate a new server-caching system.
7. Explain briefly why the F-test is useful for comparing algorithm execution times, not just their averages.

Topic 7.5: Data Visualization For Hypothesis Testing

1. Explain briefly why data visualization is important in hypothesis testing.
2. Differentiate between a bar chart and a scatter plot in terms of their best use cases.
3. Explain briefly what a box plot shows about a dataset.
4. Identify which chart type would be most appropriate for showing changes in website traffic over a month.
5. Explain briefly how visual overlap (or lack of overlap) between two groups' confidence intervals can hint at a hypothesis test's outcome.
6. State one limitation of relying only on a chart to make a "reject H_0" decision.

Topic 7.6: Ethical Issues in Data Science and Analysis

1. Define bias in the context of data collection.
2. Differentiate between sampling bias and confirmation bias, with an example of each.
3. Define "p-hacking" and explain briefly why it is considered unethical.
4. Explain briefly how running many statistical tests without correction can lead to misleading "significant" results.
5. State two practices that help prevent p-hacking.
6. Explain briefly why the Challenger Space Shuttle disaster is relevant to discussions of data ethics.
7. Explain briefly what "ethical use of data and models" involves.

Topic 7.7: Communicating Results and Conclusions

1. Explain briefly why "failing to reject H_0" is different from "proving H_0 is true."
2. State two guidelines for presenting technical statistical findings to a non-technical audience.
3. Explain briefly why a conclusion should always connect back to the original research question.
4. Identify one example of an incorrect phrase used to describe a hypothesis test result, and explain why it is incorrect.
5. Explain briefly why responsible communication of findings matters for building trust in data-driven decisions.

Section C: Long / Extensive Questions (LQs)

Topic 7.1: Introduction to Hypothesis Testing

1. Discuss in detail the concept of hypothesis testing and explain its overall role in data analysis and software engineering.
   a) Define what a hypothesis is and describe its key properties.
   b) Explain how research questions are transformed into testable hypotheses, using at least two original examples from computer science.
   c) Evaluate why hypothesis testing is considered essential for evidence-based decision-making in modern technology companies.
2. Analyze the importance of hypothesis testing as a decision-making tool in modern data-driven organizations.
   a) Discuss what could go wrong if a company made decisions purely based on intuition rather than hypothesis testing.
   b) Construct one original research question related to any area of computer science, and elaborate on how it would be transformed into a formal null and alternative hypothesis.

Topic 7.2: Types of Hypotheses

1. Elaborate in detail on the null hypothesis and alternative hypothesis, including their purposes, differences, and rules for correct formulation.
   a) Define H_0 and H_1 and explain their relationship using the courtroom analogy.
   b) Discuss the rules that must be followed when formulating mathematically sound hypotheses.
   c) Construct a complete pair of H_0 and H_1 statements for a scenario of your choice involving a mobile app feature, and justify whether it should be a one-tailed or two-tailed test.
2. Evaluate the importance of correctly formulating hypotheses before conducting any statistical test, and discuss the consequences of poorly formulated hypotheses.
   a) Discuss what can go wrong if H_0 and H_1 overlap or are not measurable.
   b) Analyze a real or hypothetical scenario where a poorly written hypothesis led to a flawed or misleading study conclusion.

Topic 7.3: Concepts in Hypothesis Testing

1. Discuss in detail the key concepts involved in hypothesis testing, including test statistics, critical regions, significance levels, and p-values.
   a) Define and explain the purpose of test statistics (z, t, F, χ^2), including when each is appropriate to use.

b) Explain the relationship between the critical region, the significance level (α), and the decision to reject or fail to reject H_0.
c) Construct a full worked example that calculates a t-value from a given sample mean, population mean, standard deviation, and sample size, and interpret the result.

2. Analyze and evaluate the concepts of Type I and Type II errors in hypothesis testing.
a) Define both types of errors and explain the trade-off between them.
b) Discuss a real-world computer science scenario (e.g., cybersecurity anomaly detection or software bug detection) where a Type I error would be costly, and another where a Type II error would be costly.
c) Evaluate how the choice of significance level (α) affects the likelihood of each type of error.

3. Construct a comprehensive explanation of the p-value, including a full step-by-step calculation and interpretation example.
a) Define what a p-value is and explain, using the coin-flip analogy, why a small p-value provides evidence against H_0.
b) Discuss why the significance level must be chosen before, not after, analyzing data.
c) Elaborate on the relationship between degrees of freedom, sample size, and the reliability of a hypothesis test's conclusion.

Topic 7.4: Performing Hypothesis Tests

1. Design and trace a complete, step-by-step hypothesis test for a real-world computer science scenario of your choice (e.g., comparing load times of two web servers).
a) State the null and alternative hypotheses.
b) Choose an appropriate significance level and justify your choice.
c) Construct a hypothetical dataset, calculate the relevant test statistic, and determine the p-value.
d) State your final decision and explain what it means for the system being tested.

2. Analyze and compare the F-test and the Chi-Square test in terms of their purposes, formulas, and appropriate use cases.
a) Explain the mathematical formula for each test and define every variable involved.
b) Construct a full worked example of an F-test comparing the variances of two algorithm execution times, including the final decision.
c) Construct a full worked example of a Chi-Square goodness-of-fit test using an observed vs. expected frequency table for a scenario involving categorical data (e.g., user preferences across 4 options).

3. Evaluate the complete 5-step hypothesis testing process and discuss why each step is essential to reaching a valid conclusion.
a) Discuss what could go wrong if any one of the five steps were skipped or performed

incorrectly.
b) Justify why data must be collected only after the hypotheses and significance level have been defined.

Topic 7.5: Data Visualization For Hypothesis Testing

1. Discuss in detail the role of data visualization in supporting and communicating hypothesis testing results.
   a) Explain the purpose and best use case of at least four different chart types (bar chart, line graph, scatter plot, box plot).
   b) Construct an example scenario where a chart could visually suggest a certain conclusion, and discuss why the actual statistical test result must still be used to confirm that conclusion.
   c) Evaluate the risks of relying solely on visualizations without performing the underlying statistical test.
2. Analyze how data visualization can both help and mislead an audience when presenting hypothesis testing results.
   a) Discuss techniques that can make visualizations clearer and more honest.
   b) Discuss techniques that could be used (unethically) to make a chart misleading, and explain how a responsible data scientist should avoid them.

Topic 7.6: Ethical Issues in Data Science and Analysis

1. Discuss in detail the major ethical issues in data science, including bias, ethical use of data, and responsible communication of findings.
   a) Define and provide an original example for each of the five types of bias discussed in this chapter (survey, sampling, gender, geographical, confirmation).
   b) Explain what p-hacking is, why it is unethical, and describe at least two practices that help prevent it.
   c) Evaluate the relationship between the Challenger Space Shuttle disaster and the broader importance of ethical, rigorous data communication in engineering.
2. Evaluate a real-world or hypothetical case study involving biased data or unethical statistical practices, and propose solutions.
   a) Construct a hypothetical scenario in which a company's dataset is biased in a way that could lead to unfair or harmful outcomes.
   b) Analyze the consequences of this bias on the affected group(s) and on the company itself.
   c) Propose a set of concrete steps the company could take to identify and correct this bias, and to communicate its findings responsibly going forward.

Topic 7.7: Communicating Results and Conclusions

1. Discuss in detail the principles of correctly interpreting and communicating hypothesis testing results.
   a) Explain the difference between "failing to reject H_0" and "proving H_0 is true," and discuss why this distinction matters.
   b) Discuss strategies for presenting technical statistical findings clearly to a non-technical audience (e.g., company stakeholders).
   c) Justify why every conclusion must be explicitly connected back to the original research question and hypotheses.

2. Design a complete, end-to-end case study that walks through a research question, hypothesis formulation, testing process, and final communicated conclusion for a computer science scenario of your choice.
   a) State the original research question and business context.
   b) Construct the null and alternative hypotheses, and describe the data collection and testing process.
   c) Present a plain-English final conclusion suitable for a non-technical audience, including a discussion of any limitations of the study.

End of Question Bank — Chapter 7: Hypothesis Testing